First Week — Overview and Matroid Basics

1 Main materials and references

We'll primarily follow Chekuri's course notes, selecting topics and adjusting the pace and chapter order to fit the group.

Other books are also good references for combinatorial optimization if the speaker would like to add something to their talk.

I've drafted a rough schedule and some content suggestions, with significant help from an LLM, and we'll use it as a guide. Of course, it's not a strict requirement; please feel free to adjust the pace and emphasis to whatever is most comfortable for the speaker.

1. Chandra Chekuri, *Combinatorial Optimization*, Spring 2022, UIUC.
   - Website: `https://courses.grainger.illinois.edu/cs586/sp2022/`
   - Course note: `https://courses.grainger.illinois.edu/cs586/sp2022/main.pdf`
2. Alexander Schrijver, *A Course in Combinatorial Optimization*.
   `https://homepages.cwi.nl/~lex/files/dict.pdf`
3. András Frank, *Connections in Combinatorial Optimization*.
   `https://andrasfrank.web.elte.hu/cikkek/FrankB1.pdf`
4. Jon Lee, *A First Course in Combinatorial Optimization*.
   `http://ndl.ethernet.edu.et/bitstream/123456789/33735/1/99.pdf.pdf`

2 Why Choose Combinatorial Optimization?

Vote on Discord, and I want to study (´ • ω • ').

3 What is Combinatorial Optimization About?

Combinatorial optimization = discrete structures + optimization.

Examples: MST, max-flow-min-cut.

Many of these problems are hard, but some become tractable if a strong structure certificate is known. Those are what we want to look into.

We skip topics that are usually covered in a basic algorithms class or graph theory course.

4 Main Topics We Hope to Cover

Hope we can cover as many of these topics in 10 weeks.

1. **Polyhedra**
   - LP
   - Integrality and TU matrices

2. **Matroids**
   - Abstraction of "greedy-friendly structures"
   - Matroid intersection / union
3. **Submodularity**
   - A function $f : 2^S \to \mathbb{N}$ is called submodular if
   $$f(A) + f(B) \geq f(A \cap B) + f(A \cup B) \qquad \forall A, B \subseteq S$$
   - Equivalently, for all $A \subseteq B \subseteq S$ and $x \in S \setminus B$,
   $$f(A \cup \{x\}) - f(A) \geq f(B \cup \{x\}) - f(B).$$
   - Examples: cut capacity, matroid rank, set cover
   - Polymatroid and Submodular flow

5 Matroid Basics

5.1 Definition of a Matroid

Definition 5.1. *A matroid $\mathcal{M}$ is a tuple $(S, \mathcal{I})$, where S is a finite ground set and $\mathcal{I} \subseteq 2^S$ is a collection of* independent sets *satisfying:*

(1) $\emptyset \in \mathcal{I}$,

(2) if $Y \subseteq X$ *and* $X \in \mathcal{I}$, *then* $Y \in \mathcal{I}$,

(3) if $X, Y \in \mathcal{I}$ *and* $|X| < |Y|$, *then there exists* $y \in Y \setminus X$ *such that* $X \cup \{y\} \in \mathcal{I}$.

5.2 Examples of Matroids

1. **Vector matroid (linear matroid).**
   $$\mathcal{M} = (S, \mathcal{I}),$$
   where
   $$S = \{v_1, v_2, \ldots, v_n\} \subseteq \mathbb{F}^m, \qquad \mathcal{I} = \{I \subseteq S : I \text{ is linearly independent}\}.$$
2. **Graphic matroid (cycle matroid).**
   Let $\mathcal{G} = (V, E)$ be an undirected multigraph (possibly with loops). Define
   $$\mathcal{M} = (E, \mathcal{I}),$$
   where
   $$\mathcal{I} = \{I \subseteq E : I \text{ induces a forest in } \mathcal{G}\}.$$
3. **Uniform matroid.**
   $$\mathcal{M} = (S, \mathcal{I}), \qquad |S| < \infty, \qquad \mathcal{I} = \{I \subseteq S : |I| \leq k\}.$$

4. **Transversal matroid.**

   Let $\mathcal{G} = (V, E)$ be a bipartite graph with bipartition V_1, V_2. Define

   $$\mathcal{I} = \{I \subseteq V_1 : \exists \text{ a matching } M \text{ in } \mathcal{G} \text{ that covers } I\}.$$

   Then

   $$\mathcal{M} = (V_1, \mathcal{I})$$

   is a matroid.

5. **Matching matroid.**

   Let $\mathcal{G} = (V, E)$ be an undirected graph. Define

   $$\mathcal{I} = \{I \subseteq V : \exists \text{ a matching } M \text{ in } \mathcal{G} \text{ that covers } I\}.$$

   Then

   $$\mathcal{M} = (V, \mathcal{I})$$

   is a matroid.

5.3 Bases, Circuits, Rank, Span, and Flats

Definition 5.2. *A set $X \subseteq S$ with $X \notin \mathcal{I}$ is called a* dependent set. *An element $x \in S$ is a* loop *if $\{x\}$ is dependent.*

Definition 5.3. *An inclusion-wise maximal independent set X is called a* basis.

Fact 5.4. *All bases have the same size, which is defined as the* rank *of S.*

Fact 5.5. *If B_1, B_2 are bases of $\mathcal{M}$ and $x \in B_1 \setminus B_2$, then there exists $y \in B_2 \setminus B_1$ such that*

$$(B_1 \setminus \{x\}) \cup \{y\}$$

is a basis of $\mathcal{M}$.

Let $\mathcal{M} = (S, \mathcal{I})$ be a matroid and let $\widehat{S} \subseteq S$. Define

$$\widehat{\mathcal{I}} = \{I \subseteq \widehat{S} : I \in \mathcal{I}\}.$$

Then

$$\widehat{\mathcal{M}} = (\widehat{S}, \widehat{\mathcal{I}})$$

is also a matroid, called the *restriction* of $\mathcal{M}$ to $\widehat{S}$.

Definition 5.6. *We say $\widehat{B}$ is a basis for $\widehat{S}$ if $\widehat{B}$ is a basis of $\widehat{\mathcal{M}}$. Its size is called the* rank *of $\widehat{S}$, denoted by*

$$r_{\mathcal{M}}(\widehat{S}).$$

Equivalently, $r_{\mathcal{M}}(\widehat{S}) = \max\{|I| : I \subseteq \widehat{S}, I \in \mathcal{I}\}$.

Fact 5.7. *Let $X \subseteq Y \subseteq S$ and let B be a basis for X. Then there exists a basis $\widehat{B} \supseteq B$ for Y.*

Proposition 5.8. *The rank function $r_{\mathcal{M}}$ is submodular: for any $X, Y \subseteq S$,*

$$r_{\mathcal{M}}(X \cup Y) + r_{\mathcal{M}}(X \cap Y) \leq r_{\mathcal{M}}(X) + r_{\mathcal{M}}(Y).$$

Proof. It is equivalent to show that for all $X \subseteq Y \subseteq S$ and $z \in S$,

$$r_{\mathcal{M}}(X \cup \{z\}) - r_{\mathcal{M}}(X) \geq r_{\mathcal{M}}(Y \cup \{z\}) - r_{\mathcal{M}}(Y).$$

If

$$r_{\mathcal{M}}(Y \cup \{z\}) = r_{\mathcal{M}}(Y) + 1,$$

then all bases of $Y \cup \{z\}$ contain z, which implies all bases of $X \cup \{z\}$ contain z, and hence

$$r_{\mathcal{M}}(X \cup \{z\}) = r_{\mathcal{M}}(X) + 1.$$

□

Definition 5.9. *An inclusion-wise minimal dependent set X is called a* circuit.

Lemma 5.10. *Let C_1, C_2 be two different circuits and let $x \in C_1 \cap C_2$. Then for any $y \in C_1 \setminus C_2$, there exists a circuit*

$$C \subseteq (C_1 \cup C_2) \setminus \{x\}$$

containing y.

Proof. First observe that $(C_1 \cup C_2) \setminus \{x\}$ contains a circuit. Otherwise, suppose $(C_1 \cup C_2) \setminus \{x\}$ is independent. Since it has size $|C_1 \cup C_2| - 1$, we would have

$$r_{\mathcal{M}}(C_1 \cup C_2) = |C_1 \cup C_2| - 1.$$

Since $C_1 \cap C_2$ is independent, we can find a basis B of $C_1 \cup C_2$ containing $C_1 \cap C_2$. Then B must contain either C_1 or C_2, contradicting the independence of B.

Let B be a basis of $C_1 \cup C_2$ containing $C_2 \setminus \{x\}$. Clearly $x \notin B$. If $y \notin B$, then $B \cup \{y\}$ contains a circuit, and we are done.

Now suppose $y \in B$. Take another basis B' of $C_1 \cup C_2$ containing $C_1 \setminus \{y\}$. Then there exists $z \in B' \setminus B$ such that

$$(B \setminus \{y\}) \cup \{z\}$$

is a basis. Note that $z \neq x$, since otherwise it would contain C_2, reducing to the first case. □

Corollary 5.11. *If $X \in \mathcal{I}$ and $y \notin X$, then either:*

1. *$X \cup \{y\} \in \mathcal{I}$, or*
2. *there is a unique circuit in $X \cup \{y\}$.*

Corollary 5.12. *If B_1, B_2 are bases of $\mathcal{M}$ and $x \in B_2 \setminus B_1$, then there exists $y \in B_1 \setminus B_2$ such that*

$$(B_1 \cup \{x\}) \setminus \{y\}$$

is a basis of $\mathcal{M}$.

Definition 5.13. *For any $X \subseteq S$, the* span *of X is defined by*

$$span_{\mathcal{M}}(X) := \{y \in S : r_{\mathcal{M}}(X \cup \{y\}) = r_{\mathcal{M}}(X)\}.$$

A set X is called spanning *if*

$$\text{span}_{\mathcal{M}}(X) = S.$$

Properties (Exercises). If $U \subseteq \text{span}_{\mathcal{M}}(T)$, then

$$\text{span}_{\mathcal{M}}(U) \subseteq \text{span}_{\mathcal{M}}(T).$$

Definition 5.14. *A set X is a* flat *of $\mathcal{M}$ if*

$$\text{span}_{\mathcal{M}}(X) = X.$$

Properties.(Exercise)

- If F_1, F_2 are flats, then so is $F_1 \cap F_2$.
- If F is a flat and $t \in S \setminus F$, let F' be the smallest flat containing $F \cup \{t\}$. Then there is no flat strictly between F and F'.
- I is independent if for any $y \in I$, there is a flat $F \supseteq I \setminus \{y\}$ but $y \notin F$.

5.4 Operations on Matroids

Definition 5.15. *Let $\mathcal{M}_i = (S_i, \mathcal{I}_i)$ be a sequence of matroids on pairwise disjoint ground sets. The* direct sum *of $\mathcal{M}_1 \cdots \mathcal{M}_k$ is the matroid $\mathcal{M} = (S, \mathcal{I})$ where*

$$S = \bigcup_i S_i \qquad \text{and} \qquad \mathcal{I} = \left\{ \bigcup I_j : I_j \in \mathcal{I}_j \right\}.$$

A matroid is separable *if it can be presented as the direct sum of some smaller summands.*

Lemma 5.16. *Let $S = S_1 \sqcup S_2$ and $\mathcal{M}_1, \mathcal{M}_2$ be the restriction of $\mathcal{M}$ on them respectively, then $\mathcal{M}$ is the direct sum of $\mathcal{M}_1$ and $\mathcal{M}_2$ if and only if*

$$r_{\mathcal{M}}(S_1) + r_{\mathcal{M}}(S_2) = r_{\mathcal{M}}(S)$$

Proof. Necessity is trivial.

Conversely, if $\mathcal{M}$ is not the direct sum, then there exist independent sets $I_1 \subseteq S_1, I_2 \subseteq S_2$ such that $I_1 \cup I_2 \notin \mathcal{I}$. Equivalently, there is a circuit C intersecting both S_1 and S_2. Implying that $r_{\mathcal{M}}(C) = |C| - 1$ and $r_{\mathcal{M}}(S_1 \cap C) + r_{\mathcal{M}}(S_2 \cap C) = |C|$.

From the submodularity of $r_{\mathcal{M}}$, it follows that $r_{\mathcal{M}}(S_i) + r_{\mathcal{M}}(C) \geq r_{\mathcal{M}}(S_i \cup C) + r_{\mathcal{M}}(S_i \cap C)$, and

$$\begin{aligned} r_{\mathcal{M}}(S) + 2r_{\mathcal{M}}(C) &= (r_{\mathcal{M}}(S_1) + r_{\mathcal{M}}(C)) + (r_{\mathcal{M}}(S_2) + r_{\mathcal{M}}(C)) \\ &\geq (r_{\mathcal{M}}(S_1 \cup C) + r_{\mathcal{M}}(S_1 \cap C)) + (r_{\mathcal{M}}(S_2 \cup C) + r_{\mathcal{M}}(S_2 \cap C)) \\ &\geq r_{\mathcal{M}}(S_1 \cup S_2 \cup C) + r_{\mathcal{M}}\big((S_1 \cup C) \cap (S_2 \cap C)\big) + r_{\mathcal{M}}(S_1 \cap C) + r_{\mathcal{M}}(S_2 \cap C) \\ &= r_{\mathcal{M}}(S) + r_{\mathcal{M}}(C) + (r_{\mathcal{M}}(C) + 1). \end{aligned}$$

a contradiction. □

Definition 5.17. *A matroid $\mathcal{M}$ is* connected *if the following equivalent conditions hold:*

1. *For each nonempty $U \subseteq S$,*

$$r_{\mathcal{M}}(U) + r_{\mathcal{M}}(S \setminus U) > r_{\mathcal{M}}(S).$$

2. *For each distinct $s, t \in S$, there is a circuit containing both s and t.*
3. *$\mathcal{M}$ is non-separable*

The definition corresponded to the biconnectivity in graph theory.

Dual

A dual matroid $\mathcal{M}^* = (S, \mathcal{I}^*)$ is defined by

$$\mathcal{I}^* := \{I \subseteq S : S \setminus I \text{ is spanning in } \mathcal{M}\},$$

that is,

$$r_{\mathcal{M}}(S \setminus I) = r_{\mathcal{M}}(S).$$

Take the graphic matroid for example, assuming G is connected:

Matroid	Independent sets	Bases	Circuits
$\mathcal{M}$	subforests	spanning trees	cycles
$\mathcal{M}^*$	complements of connected spanning subgraphs	complements of spanning trees	minimal cuts

Properties.(Exercises)

1. $\mathcal{M}^{**} = \mathcal{M}$.
2. B is a basis of $\mathcal{M}$ if and only if $S \setminus B$ is a basis of $\mathcal{M}^*$.
3. For $U \subseteq S$,
$$r_{\mathcal{M}^*}(U) = |U| + r_{\mathcal{M}}(S \setminus U) - r_{\mathcal{M}}(S).$$

Exercise 5.18. *$\mathcal{M}$ is connected if and only if $\mathcal{M}^*$ is connected.*

Deletion

Let $e \in S$. Then $\mathcal{M} \setminus e$ is a matroid, namely the restriction of $\mathcal{M}$ to $S \setminus \{e\}$. This extends to any $Z \subseteq S$.

Contraction

Let $e \in S$. The contraction with respect to e is defined by

$$\mathcal{M}/e = (\mathcal{M}^* \setminus e)^*.$$

This extends to any $Z \subseteq S$.

Another, more natural, definition:

$$\mathcal{M}/e = (S \setminus \{e\}, \mathcal{I}'), \qquad \mathcal{I}' = \{I \subseteq S \setminus \{e\} : I \cup \{e\} \in \mathcal{I}\}.$$

More generally,

$$\mathcal{M}/Z = (S \setminus Z, \mathcal{I}'),$$

where

$$\mathcal{I}' = \{I \subseteq S \setminus Z : I \cup X \in \mathcal{I}\},$$

where X is a basis of Z.

Properties.

1. For any $Y \subseteq S \setminus Z$,
$$r_{\mathcal{M}/Z}(Y) = r_{\mathcal{M}}(Y \cup Z) - r_{\mathcal{M}}(Z).$$
2. $$\mathcal{M}/\{e_1, e_2\} = (\mathcal{M}/e_1)/e_2 = (\mathcal{M}/e_2)/e_1.$$

Minor

A matroid $\mathcal{M}'$ is a *minor* of $\mathcal{M}$ if it can be obtained by a sequence of contractions and deletions.

5.5 Matroid Representation

A matroid $\mathcal{M} = (S, \mathcal{I})$ is *representable* over a field $\mathbb{F}$ if there exists an isomorphic vector matroid

$$\mathcal{M}' = (S', \mathcal{I}')$$

, that is, there is a bijection $f : S \to S'$ such that

$$A \in \mathcal{I} \iff f(A) \in \mathcal{I}'.$$

All matroids we have seen so far are representable.

For example, a graphic matroid is representable over any field by sending

$$(i, j) \mapsto e_i - e_j.$$

where e_i is the n-dimensional vector with its i-th dimension equals to 1 and 0 elsewhere.

If $\mathcal{M}$ is representable over $\mathbb{F}$, then so are its dual, deletion, and contraction.

A matroid is *binary* if it can be represented over $\mathbb{F}_2$. It is *regular* if it can be represented over every field. As we just saw, any graph matroid is regular.

The Fano matroid is an example of a matroid that is representable over $\mathbb{F}_2$ but not over $\mathbb{R}$.

The smallest *non-representable matroid* is the *Vámos matroid*. Let S be an 8-element ground set. Its independent sets are

$$\mathcal{I} = \left\{ I \subseteq S \ : \ |I| \leq 4 \text{ and } I \text{ is not one of the five rectangles in Figure 1} \right\}.$$

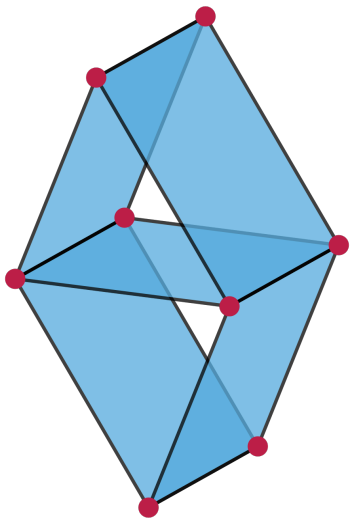


Figure 1: The Vámos matroid. [1]

[1] The eight points form four pairs; the shaded quadrilaterals are the unions of two pairs that are dependent. Five of the six such unions are dependent, but any vector representation would force the sixth to be dependent as well, so the Vámos matroid is not representable over any field. Figure from Wikipedia, "Vámos matroid".

5.6 Maximum Weight Independent Set and Greedy algorithm

Let $w : S \to \mathbb{R}_+$. Find

$$\arg\max_{I \in \mathcal{I}} w(I).$$

The greedy algorithm can be stated as follows:

1. Discard all $e \in S$ where $w(e) \leq 0$ or e is a loop.
2. Let $S = \{e_1, \ldots, e_n\}$ such that $w(e_1) \geq w(e_2) \geq \cdots \geq w(e_n)$.
3. $X \leftarrow \emptyset$.
4. For $i = 1$ to n do

   if $(X + e_i \in \mathcal{I})$ then $X \leftarrow X + e_i$.
5. Output X.

In a graph matroid, this is just Kruskal's algorithm. The proof is also the same.